Research Proposal

A Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks

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Abstract

The rapid proliferation of Internet of Things (IoT) technologies across diverse domains such as smart cities, healthcare, industrial automation, and critical infrastructure has introduced significant cybersecurity challenges. Traditional perimeter-based security models are becoming inefficient in highly distributed cross-domain IoT environments, where devices frequently interact across multiple administrative and security domains. Zero Trust Architecture (ZTA), founded on the principle of "Never Trust, Always Verify," offers a promising security paradigm but faces implementation challenges including frequent authentication requirements, trust management complexity, scalability concerns, and communication overhead.

In order to overcome such challenges, this study proposes a Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks. This framework combines Federated Learning (FL) with Zero Trust principles to provide privacy-preserving, scalable and efficient cross-domain trust evaluation. As an alternative to existing solutions that rely on centralized trust management systems suffering from single points of failure, scalability constraints and privacy concerns, the proposed framework allows multiple IoT security domains to jointly train trust models without the need to exchange raw device data.

The framework includes four main components: a Device Layer for data generation and behavioral monitoring, a Trust Management Layer for dynamic trust score generation and risk assessment, a Federated Learning Layer for local model training and global trust aggregation using a modified Federated Averaging algorithm, and a Zero Trust Control Layer for continuous authentication and authorization based on NIST SP 800-207 standards. Trust assessments are calculated on behavioral, historical and contextual attributes with adaptive weighting.

The key contributions of this work include: (1) Development of an adaptive trust management framework that supports multi-security domain operation. (2) Introduction of a trust migration mechanism enabling trust portability among heterogeneous IoT domains without the need for complete re-authentication. (3) Application of Federated Learning combined with Zero Trust trust orchestration to ensure privacy preservation. (4) Design of scalable trust coordination protocols suitable for large IoT networks.

The study uses a Design Science Research (DSR) methodology combined with quantitative experimental evaluation. Performance will be assessed using security metrics (trust

accuracy, detection rate, false positive rate), efficiency metrics (authentication latency, communication overhead, computational overhead), and scalability metrics (throughput, number of supported devices, cross-domain response time). The proposed framework is to reduce authentication latency and communication overhead while maintaining strong security guarantees, addressing critical limitations identified in existing Zero Trust and trust management solutions.
Keywords: Zero Trust Architecture, Federated Learning, Trust Management, Cross-Domain Authentication, IoT Security, Adaptive Trust Evaluation, Privacy Preservation

1 Introduction

1.1 Introduction

The fast spread of Internet of Things (IoT) technologies has transformed modern digital ecosystems by enabling seamless connectivity among billions of heterogeneous devices. IoT systems are increasingly deployed across diverse application domains, including smart cities, healthcare, industrial automation, transportation, energy management, and critical infrastructure. These systems generate enormous amounts of data and facilitate real-time decision-making, significantly improving operational efficiency and service delivery.

Despite these advantages, the widespread adoption of IoT introduces substantial cybersecurity challenges. Traditional security models, which rely on perimeter-based protection and implicit trust assumptions, are becoming ineffective in highly distributed and dynamic environments. IoT devices frequently interact across multiple administrative and security domains, creating complex trust relationships and expanding the attack surface. As a result, cyber threats such as unauthorized access, lateral movement attacks, identity spoofing, insider threats, and session hijacking have become increasingly prevalent.

To address these challenges, Zero Trust Architecture (ZTA) has emerged as a modern security paradigm founded on the principle of "Never Trust, Always Verify." Unlike conventional security approaches, ZTA continuously verifies users, devices, and applications before granting access to resources. While ZTA significantly improves security, implementing it in large-scale cross-domain IoT environments is still challenging due to frequent authentication requirements, trust management complexity, scalability concerns, and communication overhead.

Recent advances in Federated Learning (FL) offer promising opportunities for addressing these challenges. FL enables multiple entities to collaboratively train machine learning models without sharing raw data, thereby preserving privacy while supporting distributed intelligence. By integrating FL with Zero Trust principles, adaptive trust management mechanisms can be developed to facilitate efficient cross-domain authentication, dynamic trust evaluation, and scalable authorization decisions.

This research presents a Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks to enhance security, privacy, scalability, and trust coordination across heterogeneous IoT environments.

1.2 Background of the Study

Cybersecurity has moved from perimeter-based security models to identity-centric security architectures. Older methods of network security were based on the idea that all entities inside a trusted network boundary could be considered trustworthy. But the development of cloud computing, remote work, mobile devices, and IoT ecosystems has made that assumption no longer valid.

Zero Trust Architecture, as formally defined by the National Institute of Standards and Technology (NIST SP 800-207), gets rid of any assumed trust and calls for the continuous verification of each and every access request regardless of where the request comes from. This approach includes principles such as least-privilege access, continuous authentication, dynamic authorization, micro-segmentation, and continuous monitoring.

In IoT environments, devices commonly communicate over a number of domains, for example, enterprise networks, cloud platforms, industrial systems, and service provider infrastructures. Currently, the cross-domain authentication methods in use usually involve full re-authentication each time a device enters a new security domain. This process introduces significant latency and computational overhead while limiting interoperability.

Trust management has become as a critical component of ZTA, enabling security systems to assess the trustworthiness of devices and users dynamically. However, most existing trust management approaches depend on centralized architectures that face scalability limitations and privacy concerns.

Federated Learning provides a decentralized alternative by enabling collaborative trust evaluation without exposing sensitive device data. Through distributed model training and secure parameter exchange, FL can support adaptive trust assessment while preserving privacy. The integration of FL with ZTA therefore represents a promising direction for improving cross-domain trust management in large-scale IoT networks.

1.3 Problem Statement

The growing complexity of cross-domain IoT ecosystems presents significant challenges for existing Zero Trust implementations. Current authentication and authorization mechanisms often suffer from excessive communication overhead, repeated authentication processes, high latency, and limited trust portability across security domains.

Furthermore, existing trust management systems are predominantly centralized, which makes them vulnerable to single points of failure, scalability limitations, and privacy risks. While Federated Learning has demonstrated effectiveness in privacy-preserving collaborative learning, its application to adaptive trust management within cross-domain Zero Trust IoT environments remains insufficiently explored.

Consequently, there is a need for a scalable, privacy-preserving, and adaptive trust

management framework capable of dynamically evaluating trust across multiple security domains while supporting efficient authentication and authorization decisions in Zero Trust IoT networks.

1.4 Significance of the Study

This study is significant for several reasons:

It contributes to the improvement of Zero Trust security frameworks for large-scale IoT environments.

It addresses the missing research on cross-domain trust management in current literature.

It leverages Federated Learning to enhance privacy preservation by eliminating the need to exchange raw device data.

It improves authentication efficiency by enabling adaptive trust evaluation and trust migration across security domains.

It provides a scalable framework suitable for heterogeneous IoT deployments involving multiple administrative domains.

It contributes to both academic research and practical cybersecurity implementations in smart cities, healthcare systems, industrial IoT, and critical infrastructure.

1.5 Objectives of the Study

General Objective To develop a Federated Learning-Based Adaptive Trust Management Framework that enhances cross-domain authentication and authorization in Zero Trust IoT networks.

Specific Objectives To analyze existing trust management approaches in Zero Trust IoT environments.

To design a federated trust management framework capable of supporting cross-domain trust evaluation in a Zero Trust IoT environment.

To develop an adaptive trust scoring mechanism based on IoT device behavior and contextual information.

To integrate Federated Learning into the trust evaluation process while preserving privacy.

To evaluate the framework in terms of security, scalability, authentication latency, trust accuracy, and communication overhead.

To compare the proposed framework with existing Zero Trust and trust management solutions.

1.6 Research Questions

The study seeks to answer the following research questions:

RQ1: What are the limitations of existing trust management approaches in cross-domain Zero Trust IoT environments?

RQ2: How can Federated Learning be utilized to facilitate privacy-preserving trust evaluation across multiple security domains?

RQ3: How can adaptive trust scores be generated to support dynamic authentication and authorization decisions?

RQ4: To what extent can the proposed framework reduce authentication latency and communication overhead?

RQ5: How does the proposed framework compare to other existing Zero Trust trust management solutions in terms of security, scalability, and efficiency?

1.7 Limitations of the Study

This study is subject to the following limitations:

The proposed framework will be evaluated primarily through simulation and experimental environments rather than large-scale real-world deployment.

The research focuses on IoT networks operating under Zero Trust principles and may not directly address other network architectures.

The Federated Learning model assumes participating domains are cooperative and willing to exchange model parameters.

The study does not address post-quantum cryptographic requirements.

Resource-constrained IoT devices may require additional optimization mechanisms beyond the scope of this research.

1.8 Methodology of the Study

This study adopts a Design Science Research methodology combined with quantitative experimental evaluation.

The research process consists of the following phases:

Literature Review: Conduct a systematic review of Zero Trust Architecture, trust management, Federated Learning, and cross-domain authentication.

Requirement Analysis: Identify security, privacy, scalability, and trust management requirements.

Framework Design: Design the proposed federated trust management architecture and trust evaluation mechanisms.

Model Development: Develop adaptive trust scoring algorithms and federated learning processes.

Simulation and Implementation: Implement the framework using simulation tools and machine learning environments.

Performance Evaluation: Evaluate the framework using metrics such as authentication latency, trust accuracy, communication overhead, throughput, scalability, and security resilience.

Comparative Analysis: Compare the proposed framework against existing Zero Trust and trust management approaches.

1.9 Thesis Structure

This thesis is organized into six chapters as follows:

Chapter One presents the introduction, background, problem statement, significance, objectives, research questions, methodology, limitations, and overall thesis organization.

Chapter Two reviews the relevant literature on Zero Trust Architecture, IoT security, trust management systems, Federated Learning, and cross-domain authentication mechanisms.

Chapter Three presents the conceptual framework, system architecture, assumptions, and design requirements of the proposed solution.

Chapter Four describes the proposed Federated Learning-Based Adaptive Trust Management Framework, including trust evaluation algorithms and cross-domain trust management mechanisms.

Chapter Five presents the implementation details, experimental setup, simulation results, performance evaluation, and comparative analysis.

Chapter Six summarizes the research findings, contributions, limitations, and recommendations for future work.

2 Literature Review

2.1 Introduction

This chapter reviews the theoretical foundations and recent research related to Zero Trust Architecture (ZTA), Internet of Things (IoT) security, trust management systems, cross-domain authentication and authorization, and Federated Learning (FL). The chapter critically analyzes recent studies and identifies existing limitations that motivate the development of the proposed Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks.

2.2 Zero Trust Architecture (ZTA)

2.2.1 Evolution of Network Security Models

Traditional Perimeter-Based Security Models (PBSM) assume that entities inside a network boundary can be trusted. However, cloud computing, remote work, mobile devices, and IoT ecosystems have rendered perimeter-based defenses insufficient.

The concept of Zero Trust was first introduced by John Kindervag in 2010 and later standardized by NIST SP 800-207. The core principle of ZTA is "Never Trust, Always Verify."

Core Principles of ZTA Continuous Authentication
Continuous Authorization
Least Privilege Access
Micro-Segmentation
Continuous Monitoring
Dynamic Trust Evaluation

2.2.2 Benefits of ZTA in IoT Environments

ZTA addresses major IoT security challenges including:
Insider threats
Lateral movement attacks

Device impersonation

Credential theft

Unauthorized access

However, implementing ZTA in large-scale IoT environments introduces significant challenges related to scalability, trust management, and authentication overhead.

2.3 Trust Management in Zero Trust Environments

2.3.1 Concept of Trust

Trust represents the degree of confidence that a device, user, or service will behave according to security expectations.

Trust evaluation commonly utilizes:

Behavioral Trust

Historical Trust

Contextual Trust Reputation-Based Trust

2.3.2 Existing Trust Models

Existing trust models can be classified into:

Centralized Trust Models Advantages:

Simplicity

Disadvantages:

Single Point of Failure

Limited Scalability

Distributed Trust Models Advantages:

Scalability

Fault Tolerance

Disadvantages:

Complex Trust Coordination

2.3.3 Challenges of Current Trust Models

Current trust models suffer from:

Static trust calculation

Lack of trust migration
Poor cross-domain interoperability
Privacy concerns
High computational complexity

2.4 Cross-Domain Authentication and Authorization

2.4.1 Cross-Domain Security Requirements

Modern IoT systems frequently operate across:
Enterprise Networks
Cloud Environments
Industrial Control Systems
Service Provider Networks
Cross-domain interaction requires:
Identity Verification
Trust Transfer
Authorization Coordination

2.4.2 Existing Cross-Domain Authentication Approaches

One-Time Authentication Advantages:
Low authentication overhead
Limitations:
Session hijacking
Lateral movement attacks

Blockchain-Based Authentication Advantages:
Decentralization
Transparency
Limitations:
High latency
High computational overhead
Energy consumption

Zero Trust Authentication Advantages:
Continuous verification
Dynamic authorization
Limitations:

Repeated authentication costs

Trust portability limitations

2.5 Federated Learning in Cybersecurity

2.5.1 Overview of Federated Learning

Federated Learning is a decentralized machine learning paradigm where multiple participants collaboratively train a model without sharing raw data.

Benefits include:

Privacy Preservation

Reduced Data Exposure

Distributed Intelligence

2.5.2 Federated Learning for Security Applications

FL has been applied in:

Intrusion Detection Systems

Malware Detection

Anomaly Detection

Trust Evaluation

2.5.3 Challenges of FL

Model Poisoning Attacks

Data Poisoning

Byzantine Nodes

Non-IID Data Distribution

Aggregation Complexity

2.6 Review of Recent Zero Trust Research

2.6.1 Zero Trust Architecture: A Systematic Literature Review (Gambo & Almulhem, 2025)

Main Contributions Comprehensive taxonomy of ZTA applications.

Identification of enabling technologies.

Analysis of adoption challenges.

Key Findings Strong growth of ZTA adoption.
Significant challenges in trust management.
Limited cross-domain implementations.

Identified Gaps Lack of trust management frameworks.
Privacy-preserving trust evaluation remains underexplored.
Scalability issues in large-scale IoT systems.

2.6.2 Zero Trust Architecture in the Modern Threat Landscape (2026)

Main Contributions Analysis of vulnerabilities and attack surfaces.
Classification of Zero Trust weaknesses.

Key Findings Identity systems remain primary attack targets.
Policy engines become critical points of failure.
Cross-domain orchestration remains unsolved.

Identified Gaps Absence of unified trust orchestration.
Lack of standard evaluation frameworks.
Limited real-world deployments.

2.6.3 DAGA: Dynamic Authentication and Granularized Authorization (2026)

Main Contributions Cross-domain pre-authentication.
LSTM-based trust prediction.
Granularized authorization.

Key Findings Reduced authentication latency.
Improved authorization efficiency.

Limitations Trust migration remains partially addressed.
Dependence on simulation-based evaluation.
No protection against FL poisoning attacks.

Table 2.1: Comparative Analysis of Existing Studies

Study	Main Focus	Technology Used	Strengths	Limitations
Gambo & Almulhem (2025)	ZTA Review	SLR	Comprehensive taxonomy	No proposed framework
Modern Threat Landscape (2026)	Vulnerabilities	SLR	Threat classification	No implementation model
DFL-ZTA (2025)	Cross-Domain Authentication	DFL	Privacy preservation	Limited trust orchestration
DAGA (2026)	Adaptive Authentication	FL + LSTM	Reduced latency	Limited trust migration
Proposed Study	Adaptive Trust Management	FL + Dynamic Trust	Cross-domain trust orchestration	To be evaluated

2.7 Comparative Analysis of Existing Studies

2.8 Research Gap Analysis

Based on the reviewed literature, several gaps remain unresolved:

Lack of adaptive trust management frameworks capable of operating across multiple security domains.

Absence of effective trust migration mechanisms between heterogeneous IoT domains.

Limited integration between Federated Learning and Zero Trust trust orchestration.

Insufficient privacy-preserving trust evaluation techniques.

Lack of scalable trust coordination mechanisms for large-scale IoT deployments.

Existing solutions focus on authentication optimization rather than holistic trust management.

Therefore, a Federated Learning-Based Adaptive Trust Management Framework is proposed to address these limitations.

2.9 Conceptual Framework

The proposed framework is founded on the integration of:

Zero Trust Architecture

Federated Learning

Dynamic Trust Evaluation

Cross-Domain Trust Orchestration

Adaptive Authorization

These components collectively support secure, scalable, and privacy-preserving trust management across heterogeneous IoT environments.

2.10 Chapter Summary

This chapter reviewed the theoretical foundations and contemporary literature related to Zero Trust Architecture, trust management, cross-domain authentication, and Federated Learning. The review identified several limitations in existing approaches, particularly regarding adaptive trust management and cross-domain trust orchestration. These limitations establish the research gap addressed by the proposed framework. The next chapter presents the research design and system architecture of the proposed solution.

3 Research Methodology And Proposed Framework

3.1 Introduction

This chapter presents the research methodology and the proposed Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks. The chapter describes the research design, system architecture, trust management model, federated learning integration, trust evaluation mechanism, experimental environment, and performance evaluation metrics. The proposed framework is developed based on Zero Trust Architecture (ZTA) principles defined by the National Institute of Standards and Technology (NIST) and recent advancements in federated learning and trust management systems .

3.2 Research Methodology

3.2.1 Research Approach

This study adopts a Design Science Research (DSR) methodology.

Design Science Research is widely used in information systems and cybersecurity research because it focuses on the creation and evaluation of innovative artifacts designed to solve identified problems .

The DSR process consists of:

Problem Identification

Objective Definition

Framework Design

Artifact Development

Demonstration

Evaluation

Communication

The methodology is suitable because this research aims to develop and evaluate a novel adaptive trust management framework for cross-domain Zero Trust IoT environments.

3.2.2 Research Process

The research process consists of six phases:

Phase 1: Literature Review A systematic review of:
Zero Trust Architecture
Trust Management
Cross-Domain Authentication
Federated Learning
was conducted to identify existing limitations and research gaps .

Phase 2: Requirement Analysis Security and operational requirements are extracted from:
NIST SP 800-207
Cross-domain IoT authentication studies
Federated Learning security frameworks

Phase 3: Framework Design An adaptive trust management architecture is designed.

Phase 4: Model Development Trust evaluation and federated learning models are developed.

Phase 5: Simulation The framework is implemented and tested.

Phase 6: Performance Evaluation Results are analyzed and compared with baseline approaches.

3.3 System Model

3.3.1 Network Architecture

The proposed environment consists of multiple IoT security domains.
Each domain contains:
IoT Devices
Local Trust Manager
Policy Decision Point (PDP)
Policy Enforcement Point (PEP)
Federated Learning Agent
The architecture follows NIST Zero Trust Architecture recommendations by separating policy enforcement and policy decision functions .

Domains The system includes:

Domain A

Domain B

Domain C

Cloud Service Domain

Each domain maintains local trust information while participating in federated trust learning.

3.3.2 Threat Model

The proposed framework considers the following threats:

External Threats Identity Spoofing

Credential Theft

Replay Attacks

Internal Threats Insider Attacks

Lateral Movement

Session Hijacking

Federated Learning Threats Model Poisoning

Data Poisoning

Byzantine Attacks

These threats have been identified as major security concerns in recent ZTA and FL research.

3.4 Proposed Framework

3.4.1 Framework Overview

The proposed framework integrates:

Zero Trust Architecture

Federated Learning

Adaptive Trust Evaluation

Cross-Domain Trust Migration

Dynamic Authorization

The objective is to reduce authentication latency while maintaining strong security guarantees.

Unlike existing approaches that require complete re-authentication whenever a device enters a new domain, the proposed framework enables trust portability through federated trust sharing.

3.4.2 Framework Components

A. Device Layer

Responsible for Data generation
Resource requests
Behavioral monitoring
B. Trust Management Layer

Responsible for Trust score generation
Trust score updates
Risk evaluation
C. Federated Learning Layer

Responsible for Local model training
Parameter sharing
Global trust model aggregation
Based on the FL paradigm introduced by McMahan et al. (2017).
D. Zero Trust Control Layer

Responsible for Continuous Authentication
Continuous Authorization
Policy Enforcement
Following NIST SP 800-207.

3.5 Adaptive Trust Evaluation Model

3.5.1 Trust Factors

Trust scores are calculated using multiple attributes:

Behavioral Trust

Evaluates Access frequency
Access patterns
Resource usage

Historical Trust

Evaluates Previous interactions

Authentication history

Contextual Trust

Evaluates Device location

Time of access

Device state

These trust dimensions are commonly used in dynamic trust management systems .

3.5.2 Trust Score Formula

The overall trust score is calculated as:

The overall trust score is calculated as:

$$T = w_1 B + w_2 H + w_3 C \tag{3.1}$$

Where:

T = Overall Trust Score

B = Behavioral Trust

H = Historical Trust

C = Contextual Trust

w1,w2,w3 = Adaptive Weights

The weights are dynamically adjusted based on federated learning outputs.

3.6 Federated Learning-Based Trust Aggregation

3.6.1 Local Training

Each domain trains its local trust model using local device data.

Raw data never leaves the domain.

This privacy-preserving approach follows the federated learning paradigm proposed by McMahan et al.

3.6.2 Global Aggregation

The central aggregation server collects model parameters from participating domains.

A modified Federated Averaging (FedAvg) algorithm is used to produce the global trust model.

3.6.3 Cross-Domain Trust Migration

When a device moves to a new domain:

Previous trust records are requested.

Federated trust scores are evaluated.

Preliminary trust is assigned.

Continuous verification begins.

This mechanism addresses one of the major research gaps identified by Ma et al.

3.7 Authentication and Authorization Process

The proposed process consists of:

Step 1

Device Registration

Step 2

Identity Verification

Step 3

Trust Evaluation

Step 4

Risk Assessment

Step 5

Authorization Decision

Step 6

Continuous Monitoring

These stages align with Zero Trust principles described in NIST SP 800-207.

3.8 Performance Evaluation Metrics

The framework will be evaluated using:

Security Metrics Trust Accuracy

Detection Rate

False Positive Rate

Efficiency Metrics Authentication Latency

Communication Overhead

Computational Overhead

Scalability Metrics Throughput

Number of Supported Devices

Cross-Domain Response Time

These metrics are commonly adopted in DAGA and DFL-ZTA studies .

3.9 Chapter Summary

This chapter presented the research methodology and the proposed Federated Learning-Based Adaptive Trust Management Framework for Cross-Domain Zero Trust IoT Networks. The framework integrates federated learning, adaptive trust evaluation, and Zero Trust principles to enable privacy-preserving, scalable, and efficient cross-domain trust management. The next chapter will present the implementation details and experimental setup used to validate the proposed framework.

Bibliography notes

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